

Pranab Kumar

Lucknow, Uttar Pradesh

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Education

Chandigarh University

2020 – 2024

B.E. (CGPA: 7.1)

Modern Academy

2018 – 2020

Science (79%)

Skills

Languages: Python, C++, Java, JavaScript, SQL

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3, Streamlit, Material UI

Backend: Node.js, Express.js, FastAPI, REST APIs

AI: Generative AI, Large Language Models (LLMs), LangChain, CrewAI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, ChromaDB, HuggingFace, Whisper, OpenAI API, Mistral AI

Databases: MongoDB, PostgreSQL, MySQL, ChromaDB

Cloud & DevOps: Docker, Git, GitHub Actions, Azure

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, System Design

Experience

Tata Consultancy Services

Jul 2024 – Present

System Engineer (Software Engineer), Bhubaneswar

- **Designed** and developed scalable software solutions using **Python, Java, C++, React.js, Node.js, Next.js**, and **REST APIs**, delivering modular, maintainable, and high-quality enterprise applications.
- **Built** reusable backend components and API integrations using **Python, FastAPI, Node.js**, and **SQL**, improving application **scalability, maintainability**, and overall development efficiency.
- **Optimized** application performance by debugging complex production issues, improving code quality, reducing technical debt, and implementing efficient algorithms and data structures to enhance system reliability.
- **Collaborated** with cross-functional teams following **Agile Scrum** methodologies, participating in feature development, code reviews, version control using **Git**, containerization with **Docker**, and CI/CD best practices to accelerate software delivery.
- **Developed** intelligent software solutions leveraging **Generative AI, Large Language Models (LLMs), Prompt Engineering, LangChain**, and **Retrieval-Augmented Generation (RAG)** to automate workflows, improve information retrieval, and enhance developer productivity.

Coding Ninjas

2023

Teaching Assistant

- Assisted in teaching Data Structures and Algorithms; solved 400+ problems in C++ to guide students, achieving 4.5/5 average student feedback.

Projects

AI Video Assistant (RAG-based Meeting/Video Analyzer)

Python, LangChain, Whisper, ChromaDB, Mistral AI, HuggingFace, Streamlit

- Built an end-to-end AI pipeline that ingests YouTube videos or local audio/video files and auto-generates titles, summaries, action items, key decisions, and open questions using LLM-based extraction.
- Implemented local speech-to-text transcription using **Whisper**, supporting multi-language input (English/Hinglish).
- Designed a **RAG** pipeline using **LangChain, ChromaDB**, and HuggingFace embeddings to enable natural-language Q&A over full meeting/video transcripts.
- Integrated **Mistral AI** LLM via LangChain's LCEL for context-grounded conversational responses, reducing hallucinated answers by restricting retrieval to transcript-only context.
- Built an interactive **Streamlit** UI with tabbed views (Summary, Transcript, Chat) enabling non-technical users to explore and query processed video content in real time.

Multi-Agent AI Research System

Python, CrewAI, LangChain, FastAPI, OpenAI API, Docker

- **Architected** a scalable **Multi-Agent AI System** using **CrewAI** and **LangChain**, enabling autonomous collaboration between specialized **LLM** agents for intelligent research and task execution.
- **Integrated REST APIs**, external search tools, and **OpenAI API** to automate information retrieval, reasoning, and structured report generation, reducing manual research effort by **70%**.
- **Optimized Prompt Engineering**, agent orchestration, and modular **Python** workflows, improving scalability, maintainability, and end-to-end execution efficiency.

Achievements

- **Secured a Rank of 2000** among **35,000+ participants** in a **LeetCode** programming contest, demonstrating strong competitive programming and problem-solving skills.
- **Solved 700+ Data Structures & Algorithms problems** across coding platforms, including **500+ problems on LeetCode**, strengthening expertise in algorithms, data structures, and coding interviews.
- Earned a **4-Star** rating on **HackerRank** and a **3-Star** rating on **CodeChef**.

Certifications

- Data Structures in C++ Excellence Certificate | AWS Cloud – Coursera | IBM watsonx.ai Technical Essentials | AWS Cloud Technical Essentials | Full stack Development